Lean Canvas · FasalMitra

Smart crop and fertilizer advisor for small farmers

SDG 2 · Zero Hunger (also SDG 12 and SDG 13)
Shreyash Surya Dubey · Internship ID: IBMUEML007139
AICTE | IBM SkillsBuild ML & Applied AI Internship 2026

1 · PROBLEM

- Crop and fertilizer choices rely on guesswork or habit.
- Over-fertilizing wastes money and harms soil and water; the wrong crop lowers income.
- Soil Health Card values (N, P, K, pH) are hard for farmers to turn into a decision.

EXISTING ALTERNATIVES

- Soil Health Card: values, but no advice
- Extension visits: limited reach
- Advice from neighbours or input dealers, who may favour selling more

4 · SOLUTION

- Enter N, P, K, pH and local weather
- ML model ranks the top 3 crops with a suitability %
- Shows why: each value vs the crop's normal range
- Fertilizer plan: add, skip, or save
- English and Hindi web app for phones

8 · KEY METRICS

- Farmers advised
- Fertilizer (kg/ha) not applied after "skip" advice
- Share of plans an agri officer agrees with
- Repeat use next season
- Model accuracy on local data

3 · UNIQUE VALUE PROPOSITION

A soil-test card turned into a crop and fertilizer plan, in plain Hindi or English, with the reasons shown.

Unlike a generic calculator or a dealer's advice, every recommendation is explained and no input brand pays to influence it.

HIGH-LEVEL CONCEPT

A "soil-card translator" with a crop advisor built in.

9 · UNFAIR ADVANTAGE

- Unbiased: no fertilizer-brand sponsorship
- Explains every recommendation
- Retrains on any district's data in minutes
- Hindi and English from day one

5 · CHANNELS

- KVKs and extension workers
- Farmer groups and WhatsApp sharing
- College agri clubs, student ambassadors
- QR codes at soil-testing labs

2 · CUSTOMER SEGMENTS

- Small and marginal farmers
- Farmer producer organisations
- KVKs and extension workers
- NGO and CSR agriculture programmes

EARLY ADOPTERS

Farmer groups and villages reachable through KVKs, NGOs and college agri clubs.

7 · COST STRUCTURE

- Hosting and maintenance
- Translation and regional-language support
- Field validation with agronomists
- Local data updates and model retraining
- User support and training

6 · REVENUE STREAMS

- Free for farmers
- Licences for NGOs, producer organisations and CSR programmes
- Partnerships and grants with agriculture departments
- No fertilizer-brand sponsorship, so advice stays unbiased